LEMMA BLOWUP IN ALGEBRAIZATION.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebraization.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the corrected band-I follow-on screen (line C) of the extractor's stacks-candidates line: its author label is absent from the live master in every form searched and it belongs to none of the earlier judged pools. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:algebraization.tex:lemma-blowup.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original reference 0EHC: sources/algebraization.tex, source lines 4983–4983. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original equation 0EIX: sources/algebraization.tex, source lines 5589–5589. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIY: sources/algebraization.tex, source lines 5622–5622. Navigation pointer used in the counted span; the span contains the

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Source reference. Original lemma 0887: sources/coherent.tex, source lines 6631–6631. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EJF: sources/algebraization.tex, source lines 6952–6952. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0804: sources/divisors.tex, source lines 8322–8322. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07NV: sources/more-algebra.tex, source lines 10624–10624. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02IJ: sources/algebra.tex, source lines 27520–27520. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BJN: sources/dualizing.tex, source lines 4191–4191. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0AWY: sources/dualizing.tex, source lines 4235–4235. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 0BIR: sources/more-algebra.tex, source lines 12814–12814. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0338: sources/algebra.tex, source lines 46303–46303. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original proposition 0AW6: sources/more-algebra.tex, source lines 32484–32484. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIZ: sources/algebraization.tex, source lines 5760–5760. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EJI: sources/algebraization.tex, source lines 7129–7129. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIR: sources/algebraization.tex, source lines 5249–5249. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIS: sources/algebraization.tex, source lines 5297–5297. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0885: sources/coherent.tex, source lines 6812–6812. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0889: sources/coherent.tex, source lines 6487–6487. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 088B: sources/coherent.tex, source lines 7025–7025. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIU: sources/algebraization.tex, source lines 5458–5458. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EIN: sources/algebraization.tex, source lines 5171–5171. Navigation pointer used in the counted span; the span contains the

pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BFR: sources/dualizing.tex, source lines 4126–4126. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0805: sources/divisors.tex, source lines 8351–8351. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *In Situation 0EHC let $(\mathcal{F}_n)$ be an object of $Coh(U, I\mathcal{O}_U)$. Assume*

(1) *A has a dualizing complex,*
(2) *all fibres of the blowing up $b : X' \to X$ of I have dimension $\leq d-1$,*
(3) *one of the following is true*
   (a) *$(\mathcal{F}_n)$ satisfies the $(d+1, d+2)$-inequalities (Definition 0EJ3), or*
   (b) *for $y \in U \cap Y$ and a prime $\mathfrak{p} \subset \mathcal{O}^\wedge_{X,y}$ with $\mathfrak{p} \notin V(I\mathcal{O}^\wedge_{X,y})$ we have*

$$depth((\mathcal{F}^\wedge_y)_\mathfrak{p}) + \dim(\mathcal{O}^\wedge_{X,y}/\mathfrak{p}) + \delta^Y_Z(y) > d + 2$$

Then $(\mathcal{F}_n)$ extends to X.

Proof. Let $Y' \subset X'$ be the exceptional divisor. Let $Z' \subset Y'$ be the inverse image of $Z \subset Y$. Then $U' = X' \setminus Z'$ is the inverse image of U. With $\delta^{Y'}_{Z'}$ as in (0EIX) we set

$$T' = \{y' \in Y' \mid \delta^{Y'}_{Z'}(y') = 1\} \subset T = \{y' \in Y' \mid \delta^{Y'}_{Z'}(y') = 1 \text{ or } 2\}$$

By Lemma 0EIY parts (1) and (2) these are specialization stable subsets of $U' \cap Y' = Y' \setminus Z'$. Consider the object $(b|^*_{U'}\mathcal{F}_n)$ of $Coh(U', I\mathcal{O}_{U'})$, see Cohomology of Schemes, Lemma 0887. For $y' \in U' \cap Y'$ let us denote

$$\mathcal{F}^\wedge_{y'} = \lim(b|^*_{U'}\mathcal{F}_n)_{y'}$$

the "stalk" of this pullback at y'. We claim that conditions (a), (b), (c), (d), and (e) of Lemma 0EJF hold for the object $(b|^*_{U'}\mathcal{F}_n)$ on U' with d replaced by 1 and the subsets $T' \subset T \subset U' \cap Y'$. Condition (a) holds because Y' is an effective Cartier divisor and hence locally cut out by 1 equation. Condition (e) holds by Lemma 0EIY part (7). To prove (b), (c), and (d) we need some preparation.

Let $y' \in U' \cap Y'$ and let $\mathfrak{p}' \subset \mathcal{O}^\wedge_{X',y'}$ be a prime ideal not contained in $V(I\mathcal{O}^\wedge_{X',y'})$. Denote $y = b(y') \in U \cap Y$. Choose $f \in I$ such that y' is contained in the spectrum of the affine blowup algebra $A[\frac{I}{f}]$, see Divisors, Lemma 0804. For any A-algebra B denote $B' = B[\frac{IB}{f}]$ the corresponding affine blowup algebra. Denote I-adic completion by $^\wedge$. By our choice of f we get a ring map $(\mathcal{O}^\wedge_{X,y})' \to \mathcal{O}^\wedge_{X',y'}$. If we let $\mathfrak{q}' \subset (\mathcal{O}^\wedge_{X,y})'$ be the inverse image of $\mathfrak{m}^\wedge_{y'}$, then we see that $((\mathcal{O}^\wedge_{X,y})'_{\mathfrak{q}'})^\wedge = \mathcal{O}^\wedge_{X',y'}$. Let $\mathfrak{p} \subset \mathcal{O}^\wedge_{X,y}$ be the corresponding prime. At this point we have a commutative

diagram

$$\begin{array}{ccccccc}
\mathcal{O}^\wedge_{X,y} & \longrightarrow & (\mathcal{O}^\wedge_{X,y})' & \longrightarrow & (\mathcal{O}^\wedge_{X,y})'_{\mathfrak{q}'} & \xrightarrow[\beta]{} & \mathcal{O}^\wedge_{X',y'} \\
\downarrow & & \downarrow{\scriptstyle\alpha} & & \downarrow & & \downarrow \\
\mathcal{O}^\wedge_{X,y}/\mathfrak{p} & \longrightarrow & (\mathcal{O}^\wedge_{X,y}/\mathfrak{p})' & \longrightarrow & (\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'} & \xrightarrow{\gamma} & ((\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'})^\wedge \\
 & & & & & & \downarrow \\
 & & & & & & \mathcal{O}^\wedge_{X',y'}/\mathfrak{p}'
\end{array}$$

whose vertical arrows are surjective. By More on Algebra, Lemma 07NV and the dimension formula (Algebra, Lemma 02IJ) we have

$$\dim(((\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'})^\wedge) = \dim((\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'}) = \dim(\mathcal{O}^\wedge_{X,y}/\mathfrak{p}) - \operatorname{trdeg}(\kappa(y')/\kappa(y))$$

Tracing through the definitions of pullbacks, stalks, localizations, and completions we find

$$(\mathcal{F}^\wedge_y)_{\mathfrak{p}} \otimes_{(\mathcal{O}^\wedge_{X,y})_{\mathfrak{p}}} (\mathcal{O}^\wedge_{X',y'})_{\mathfrak{p}'} = (\mathcal{F}^\wedge_{y'})_{\mathfrak{p}'}$$

Details omitted. The ring maps β and γ in the diagram are flat with Gorenstein (hence Cohen-Macaulay) fibres, as these are completions of rings having a dualizing complex. See Dualizing Complexes, Lemmas 0BJN and 0AWY and the discussion in More on Algebra, Section 0BIR. Observe that $(\mathcal{O}^\wedge_{X,y})_{\mathfrak{p}} = (\mathcal{O}^\wedge_{X,y})'_{\tilde{\mathfrak{p}}}$ where $\tilde{\mathfrak{p}}$ is the kernel of α in the diagram. On the other hand, $(\mathcal{O}^\wedge_{X,y})'_{\tilde{\mathfrak{p}}} \to (\mathcal{O}^\wedge_{X',y'})_{\mathfrak{p}'}$ is flat with CM fibres by the above. Whence $(\mathcal{O}^\wedge_{X,y})_{\mathfrak{p}} \to (\mathcal{O}^\wedge_{X',y'})_{\mathfrak{p}'}$ is flat with CM fibres. Using Algebra, Lemma 0338 we see that

$$\operatorname{depth}((\mathcal{F}^\wedge_{y'})_{\mathfrak{p}'}) = \operatorname{depth}((\mathcal{F}^\wedge_y)_{\mathfrak{p}}) + \dim(F_{\mathfrak{r}})$$

where F is the generic formal fibre of $(\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'}$ and $\mathfrak{r}$ is the prime corresponding to $\mathfrak{p}'$. Since $(\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'}$ is a universally catenary local domain, its I-adic completion is equidimensional and (universally) catenary by Ratliff's theorem (More on Algebra, Proposition 0AW6). It then follows that

$$\dim(((\mathcal{O}^\wedge_{X,y}/\mathfrak{p})'_{\mathfrak{q}'})^\wedge) = \dim(F_{\mathfrak{r}}) + \dim(\mathcal{O}^\wedge_{X',y'}/\mathfrak{p}')$$

Combined with Lemma 0EIZ we get

$$\begin{aligned}
&\operatorname{depth}((\mathcal{F}^\wedge_{y'})_{\mathfrak{p}'}) + \delta^{Y'}_{Z'}(y') \\
&= \operatorname{depth}((\mathcal{F}^\wedge_y)_{\mathfrak{p}}) + \dim(F_{\mathfrak{r}}) + \delta^{Y'}_{Z'}(y') \\
&\geq \operatorname{depth}((\mathcal{F}^\wedge_y)_{\mathfrak{p}}) + \delta^{Y}_{Z}(y) + \dim(F_{\mathfrak{r}}) + \operatorname{trdeg}(\kappa(y')/\kappa(y)) - (d-1) \\
&= \operatorname{depth}((\mathcal{F}^\wedge_y)_{\mathfrak{p}}) + \delta^{Y}_{Z}(y) - (d-1) + \dim(\mathcal{O}^\wedge_{X,y}/\mathfrak{p}) - \dim(\mathcal{O}^\wedge_{X',y'}/\mathfrak{p}')
\end{aligned} \tag{0.1.1}$$

Please keep in mind that $\dim(\mathcal{O}^\wedge_{X,y}/\mathfrak{p}) \geq \dim(\mathcal{O}^\wedge_{X',y'}/\mathfrak{p}')$. Rewriting this we get

$$\begin{aligned}
&\operatorname{depth}((\mathcal{F}^\wedge_{y'})_{\mathfrak{p}'}) + \dim(\mathcal{O}^\wedge_{X',y'}/\mathfrak{p}') + \delta^{Y'}_{Z'}(y') \\
&\geq \operatorname{depth}((\mathcal{F}^\wedge_y)_{\mathfrak{p}}) + \dim(\mathcal{O}^\wedge_{X,y}/\mathfrak{p}) + \delta^{Y}_{Z}(y) - (d-1)
\end{aligned} \tag{0.1.2}$$

This inequality will allow us to check the remaining conditions.

Conditions (b) and (d) of Lemma 0EJF. Assume $V(\mathfrak{p}') \cap V(I\mathcal{O}^\wedge_{X',y'}) = \{\mathfrak{m}^\wedge_{y'}\}$. This implies that $\dim(\mathcal{O}^\wedge_{X',y'}/\mathfrak{p}') = 1$ because Z' is an effective Cartier divisor. The combination of (b) and (d) is equivalent with

$$\operatorname{depth}((\mathcal{F}^\wedge_{y'})_{\mathfrak{p}'}) + \delta^{Y'}_{Z'}(y') > 2$$

If $(\mathcal{F}_n)$ satisfies the inequalities in (3)(b) then we immediately conclude this is true by applying (0.1.2). If $(\mathcal{F}_n)$ satisfies (3)(a), i.e., the $(d+1, d+2)$-inequalities, then we see that in any case

$$\text{depth}((\mathcal{F}_y^\wedge)_\mathfrak{p}) + \delta_Z^Y(y) \geq d+1 \quad \text{or} \quad \text{depth}((\mathcal{F}_y^\wedge)_\mathfrak{p}) + \dim(\mathcal{O}_{X,y}^\wedge/\mathfrak{p}) + \delta_Z^Y(y) > d+2$$

Looking at (0.1.1) and (0.1.2) above this gives what we want except possibly if $\dim(\mathcal{O}_{X,y}^\wedge/\mathfrak{p}) = 1$. However, if $\dim(\mathcal{O}_{X,y}^\wedge/\mathfrak{p}) = 1$, then we have $V(\mathfrak{p}) \cap V(I\mathcal{O}_{X,y}^\wedge) = \{\mathfrak{m}_y^\wedge\}$ and we see that actually

$$\text{depth}((\mathcal{F}_y^\wedge)_\mathfrak{p}) + \delta_Z^Y(y) > d+1$$

as $(\mathcal{F}_n)$ satisfies the $(d+1, d+2)$-inequalities and we conclude again.

Condition (c) of Lemma 0EJF. Assume $V(\mathfrak{p}') \cap V(I\mathcal{O}_{X',y'}^\wedge) \neq \{\mathfrak{m}_{y'}^\wedge\}$. Then condition (c) is equivalent to

$$\text{depth}((\mathcal{F}_{y'}^\wedge)_{\mathfrak{p}'}) + \delta_{Z'}^{Y'}(y') \geq 2 \quad \text{or} \quad \text{depth}((\mathcal{F}_{y'}^\wedge)_{\mathfrak{p}'}) + \dim(\mathcal{O}_{X',y'}^\wedge/\mathfrak{p}') + \delta_{Z'}^{Y'}(y') > 3$$

If $(\mathcal{F}_n)$ satisfies the inequalities in (3)(b) then we see the second of the two displayed inequalities holds true by applying (0.1.2). If $(\mathcal{F}_n)$ satisfies (3)(a), i.e., the $(d+1, d+2)$-inequalities, then this follows immediately from (0.1.1) and (0.1.2). This finishes the proof of our claim.

Choose $(b|_{U'}^*\mathcal{F}_n) \to (\mathcal{F}_n'')$ and $(\mathcal{H}_n)$ in $Coh(U', I\mathcal{O}_{U'})$ as in Lemma 0EJF. For any affine open $W \subset X'$ observe that $\delta_{W\cap Z'}^{W\cap Y'}(y') \geq \delta_{Z'}^{Y'}(y')$ by Lemma 0EIY part (6). Hence we see that $(\mathcal{H}_n|_W)$ satisfies the assumptions of Lemma 0EJI. Thus $(\mathcal{H}_n|_W)$ extends canonically to W. Let $(\mathcal{G}_{W,n})$ in $Coh(W, I\mathcal{O}_W)$ be the canonical extension as in Lemma 0EIR. By Lemma 0EIS we see that for $W' \subset W$ there is a unique isomorphism

$$(\mathcal{G}_{W,n}|_{W'}) \longrightarrow (\mathcal{G}_{W',n})$$

compatible with the given isomorphisms $(\mathcal{G}_{W,n}|_{W\cap U}) \cong (\mathcal{H}_n|_{W\cap U})$. We conclude that there exists an object $(\mathcal{G}_n)$ of $Coh(X', I\mathcal{O}_{X'})$ whose restriction to U is isomorphic to $(\mathcal{H}_n)$.

If A is I-adically complete we can finish the proof as follows. By Grothendieck's existence theorem (Cohomology of Schemes, Lemma 0885) we see that $(\mathcal{G}_n)$ is the completion of a coherent $\mathcal{O}_{X'}$-module. Then by Cohomology of Schemes, Lemma 0889 we see that $(b|_{U'}^*\mathcal{F}_n)$ is the completion of a coherent $\mathcal{O}_{U'}$-module $\mathcal{F}'$. By Cohomology of Schemes, Lemma 088B we see that there is a map

$$(\mathcal{F}_n) \longrightarrow ((b|_{U'})_*\mathcal{F}')^\wedge$$

whose kernel and cokernel is annihilated by a power of I. Then finally, we win by applying Lemma 0EIU.

If A is not complete, then, before starting the proof, we may replace A by its completion, see Lemma 0EIN. After completion the assumptions still hold: this is immediate for condition (3), follows from Dualizing Complexes, Lemma 0BFR for condition (1), and from Divisors, Lemma 0805 for condition (2). Thus the complete case implies the general case. □